🌰 Seeds: Reading

Horticulture 1 · Unit 6 · Level 3 (closer to the class work) · goes with Unit 6 Definitions, Parts of a Seed, Seeds Worksheet/Quiz, Seed Tri-Fold

Name: _____

From Seed to Seedling

Propagation is making new plants. The most common way is from seed. A seed has three main parts: the seed coat, a hard covering that protects it; the embryo, the tiny plant that will grow; and the endosperm, stored food that feeds the embryo as it sprouts. In some plants, such as beans, the food is stored in the cotyledons.

Germination begins when a seed takes in water. Seeds also need warmth and air, and some tiny seeds, like petunias, need light. The root (radicle) grows down first, then the shoot pushes up. The first leaves to appear are cotyledons, or seed leaves; monocots have one and dicots have two. The leaves that come next, which look like the adult plant's leaves, are called true leaves.

Some seeds need special treatment before they will germinate. Seeds with very hard coats can be scarified, or nicked and sanded, so water can get in. Others are soaked, or chilled for weeks in a process called cold stratification, which imitates winter.

Growers use two seeding methods. Direct seeding means sowing seeds where the plant will grow, which works well for beans, carrots and corn. Indirect seeding means starting seeds in trays in a greenhouse and transplanting them later, which gives plants like tomatoes and petunias an early start. Seeds should be planted about two to three times as deep as they are wide.

Check your reading

1. Name the three main parts of a seed and the job of each.

2. What is the difference between cotyledons and true leaves?

3. When would a grower choose indirect seeding?